DRL Midsem 21-Dec-2025

Question 1

A mobile health-monitoring wristband must choose among three modes each hour, based on the current activity *(resting, moderate, high)*:

- **Mode A:** low power, moderate accuracy
- **Mode B:** medium power, high accuracy
- **Mode C:** high power, very high accuracy

The immediate benefit (reward) from each mode changes based on the user's physical activity.

a)State **why Reinforcement Learning (not supervised learning)** fits this setting, and give a **one-line distinction between immediate reward and long-term value**.
[1.5 Marks]

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b) Using the same wristband scenario:

Case 1:
Initially, assume the user's activity pattern is **independent of the modes chosen**, i.e., choosing Mode A or B today does **not affect future rewards**.

Case 2:
Later, a firmware update introduces an adaptation rule:
If the wristband samples in Mode C for too long, the battery temperature rises, reducing future rewards for power-hungry modes.

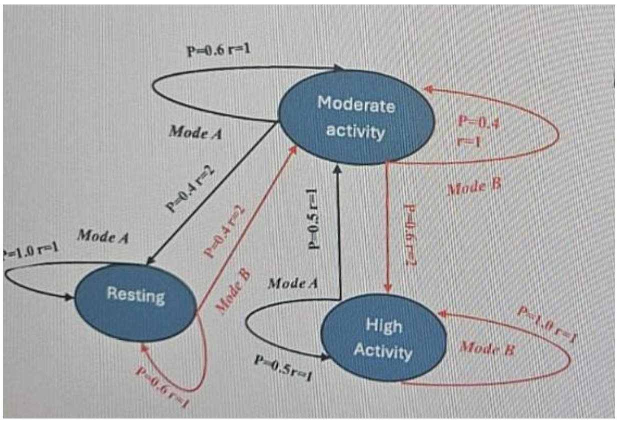
i) Classify **Case 1** as a **Multi-Armed Bandit (MAB)** or a **Finite MDP**, with one justification.
[1 Mark]

ii) Classify **Case 2** again and justify — **does the added dependency change the problem class?**
[1 Mark]

c)For the MDP given below, calculate the **Values of states** = {resting, moderate activity, high activity} (in the same order), using **synchronous Value Iteration**.

- Actions = {Mode A, Mode B} are allowed from each state
- Discount factor: **γ = 0.1**
- Solve for **1 iteration only**

[4 Marks]



Question2 :

In an **ICU environment**, clinicians must periodically adjust a patient's ventilator settings to maintain optimal blood oxygen saturation (**SpO_2**). The system can be modeled as a **finite MDP** where the patient's oxygenation levels and respiratory indicators are defined into categories such as **$lowO_2$**, **$highO_2$**, or **$OptimalO_2$**.

The clinician-assist AI can **increase pressure**, **decrease pressure**, or **maintain settings**. With **0.4 probability**, increasing or decreasing the pressure does not change a patient's health, whereas maintaining the same settings always seems to retain the patient condition as is.

The goal is to stabilize the patient's SpO_2 within safe clinical thresholds while avoiding over-ventilation. Once for **four consecutive recordings**, the SpO_2 is observed to be **Optimal**, then the patient stabilizes.

(a)

Formulate the **MDP**. State the components clearly. Diagrammatically represent the model dynamics.
Note: Assume equal likelihood for scenarios that are not explicitly mentioned in the above use case.
[2.5 Marks]

(b)

Given below reward designs, analyse the impact on the patient's safety in the given setup. Which one do you prefer and why? Justify your answer in **exactly two statements** with respect to:
i) What behavior each encourages
ii) In which scenario one design is preferable over the other

[2 Marks]

- **Design A:** For moderating from *$lowO_2$* to *$OptimalO_2$*, reward = **+10** is set.
- **Design B:** For moderating from *$OptimalO_2$* to *$HighO_2$*, reward = **−70** is set.

(c)

Is the given use case **episodic or continuous task**? Justify your answer in **no more than 30 words**.
[1.5 Marks]

(d)

If **γ = 0.01** is used, what is the impact on the patient's stability? Justify your answer in **no more than 30 words**.
[1.5 Marks]

Question 3

You can write your answers in the provided space or write your answer on a piece of paper and scan and upload the handwritten answers using the QR Code available in the **Scan and Upload Section** of this exam.

Kindly ensure all answer sheets to be uploaded against the corresponding questions only. All sheets should not be uploaded against one question.

In an **online pharma platform**, a **multi-armed bandit** is implemented that recommends **adherence-support interventions** for patients (frequent online customers) based on their current state (e.g., health conditions, stock of medication, risk of the ordered medicines, etc.) to improve quality adherence and customer retention.

Few observed interventions recommended for patients (customers) in a digital health platform include:

- **S**: SMS reminder to refill medications
- **T**: Teleconsultation scheduling with a doctor
- **R**: Reminder about lab tests
- **C**: Counseling call by pharmacist

Analyze the below observations in the given order and answer the following questions sequentially:

(1, S, 7), (2, T, 5), (3, R, 6), (4, C, 4), (5, S, 8),
(6, T, 6), (7, R, 7), (8, C, 5)

(a)

Estimate the **best intervention** based on the listed observations.
Use **exponential recency-weighted average** based value update only with **α = 0.5** to revise the **Q(actions)** for all the actions.
Assume all the values are **zero for initialization**.

[3 Marks]

(b) Refer to the results of part (a) and illustrate the significance of α**.**
What happens if α = 1 **is set instead?**
Justify answers in no more than 30 words**, using the given above use case for illustration.**

[1.5 Marks]

(c)

What is the significance of **confidence level in the UCB-based selection**?
Justify answers in **no more than 30 words** using the given above use case for illustration.

[1 Mark]

(d) Analyze only until the 5th time step**.**
On some of these time steps the ε-case may have occurred**, causing an action to have been selected at random.**

i) On which time step(s) did this **definitely occur**? Why?
[1 Mark]

ii) On which time step(s) could this **possibly have occurred**? Why?
[1 Mark]

Question 4

(a)

In a **model-free MDP**, explain why estimating **only the state value vπ(s)v_\pi(s)vπ(s)** is insufficient for selecting actions during control, even if vπv_\pivπ is estimated perfectly.
*(Ensure your answer is given in **2–3 well-articulated statements**; vague answers will be penalised.)*
[2 Marks]

(b)

In a **deterministic policy** that always selects the same action in a state, explain in **2–3 sentences** why **first-visit Monte Carlo estimation** may fail to reach an optimal policy.
Suggest **one precise way** to address this.
[2 Marks]

(c)

A **chatbot agent** operates in two states:

- s0 : user **engaged**
- s1: user **disengaged**

It may take three actions:

- a1: ask an open-ended question
- a2: recommend a resource
- a3: send a fun fact

The present **Q-values** and the **ε-soft policy** are as below:

Q-values

State	a_1	a_2	a_3
s_0	1	0.5	0
s_1	0	0.5	1

Policy

State	a_1	a_2	a_3
s_0	0.5	0.3	0.2
s_1	0.2	0.5	0.3

Now consider an episode:

(s0, a1, r = 2, s0) -> (s0, a3, r = 0, s1) -> (s1, a2, r = 3, s1) -> (s1, a2, r = -1, terminal)

Assuming **γ = 0.8** and computing returns using the **discounted-sum formulation** as per **first-visit Monte Carlo method**, produce:

- the **revised value-function table**
- the **updated policy table**

at the end of this episode, **clearly highlighting all changes**.
Show the computation corresponding to each change.

[3.5 Marks]

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